INCH-POUND
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PERFORMANCE SPECIFICATION

OIL COOLER ASSEMBLY: ENGINE AND TRANSMISSION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of oil cooler assembly for cooling engine and transmission oils (see 6.1) and is comprised of two engine oil cooler core assemblies and one transmission oil cooler assembly.

1.2 Classification. The engine and oil cooler assembly for acquisition purposes comes in three configurations as specified (see 6.2):

Configuration A - Oil cooler assembly (Drawing 8690239)
Configuration B - Core assembly, engine oil (Drawing 7356770)
Configuration C - Core assembly, transmission (torque) oil (Drawing 7356769)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2930
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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

FEDERAL

FED-STD-595 - Colors Used in Government Procurement.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

7356769	- Assembly, Torque (Transmission) Oil Cooler Core.
7356770	- Core Assembly, Engine Oil Cooler.
8690239	- Oil Cooler Assembly (Commercial).

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted).

(Copies of ANSI documents are available from the American National Standard Institute, 11 West 42nd Street, New York, NY 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B117 - Standard Practice for Operating Salt Spray (Fog) Apparatus (DoD Adopted).

(Copies of ASTM documents are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a first article sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Unless otherwise specified herein or specified in the applicable drawings, materials used shall be in accordance with the manufacturer's materials specifications for oil cooler assemblies. The materials shall be capable of meeting all the operational and environmental requirements specified herein (see 4.5.1). Recovered materials shall be used to the maximum extent practicable. Materials and special treatments shall be utilized to prevent corrosion and to resist harmful effects of dust, water and fungi. Dissimilar metals shall not be used intimate contact without protection from galvanic corrosion.

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3.3 Design and construction. The oil cooler assembly and core assemblies shall conform to all interface and envelope dimensions on Drawings 8690239, 7356770, 7356769 and listed parts, as applicable (see 4.5.1 and 4.5.2).

3.3.1 Corrosion and flux control. All core assemblies shall be internally and externally treated to remove all corrosion and excess flux. This treatment shall be followed by a rinsing and neutralizing treatment, and the core protected from further contamination during subsequent operations (see 4.5.1 and 4.5.2).

3.3.2 Internal preservative treatment. Following the operation to remove corrosion and other contaminants, the core shall be flushed with grade 10W preservative oil in a manner which will insure complete internal coverage. After preservative oil has drained, openings shall be sealed (see 4.5.1 and 4.5.2).

3.3.3 Pretreatment, priming and painting. Core assemblies shall be cleaned, pretreated and primed. Paint color shall match 16187 of FED-STD-595 (see 4.5.2).

3.3.4 Interchangeability. All parts of the oil cooler assemblies shall be interchangeable with parts of other manufacturers having the same military part number (see 4.5.2).

3.4 Performance. All assembly units shall conform to the requirements specified in 3.4.1 through 3.4.5 (see 4.5.4).

3.4.1 Leakage. When tested as specified in 4.5.4.1 and 4.5.4.2, the assembly unit shall show no evidence of leakage under cyclic and noncyclic conditions.

3.4.2 Hydrostatic pressure. When tested as specified in 4.5.4.3, the assembly unit shall show no evidence of leakage.

3.4.3 Heat rejection. Heat rejection shall be determined with air flowing through all three cores and oil flowing through the applicable arrangement as shown in figure 1 (see 4.5.4.4).

3.4.3.1 Heat rejection, core arrangement A. When tested as specified in 4.5.4.4.1, heat rejection, air pressure drop, and oil pressure drop shall conform to performance curves shown on figures 2, 5, and 8. Heat rejection shall be not more than 10 percent (%) below capacity curve. Air and oil pressure drop shall not be more than 10% above capacity curve.

3.4.3.2 Heat rejection, core arrangement B. When tested as specified in 4.5.4.4.2, heat rejection, air pressure drop, and oil pressure drop shall conform to performance curves shown on figures 3, 6, and 8. Heat rejection shall be not more than 10% below capacity curve. Air and oil pressure drop shall be not more then 10% above capacity curve.

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3.4.3.3 Heat rejection, core arrangement C. When tested as specified in 4.5.4.4.3, heat rejection, air pressure drop, and oil pressure drop shall conform to performance curves shown on figures 4, 7, and 8. Heat rejection shall be not more than 10% below capacity curve. Air and oil pressure drop shall be not more than 10% above capacity curve.

3.4.4 Vibration. When tested as specified in 4.5.4.5, the assembly shall show no evidence of seepage or leakage.

3.4.5 Corrosion resistance. When tested as specified in 4.5.4.6, the external surfaces of the assembly unit shall show no evidence of corrosion that would affect performance in any manner.

3.5 Marking. Each oil cooler assembly and core assembly shall be marked with permanent and legible characters and shall be made of a material suitable to withstand the environmental requirements specified herein (see 4.5.2). The assembly shall be marked with the National Stock Number (NSN) the part or identifying number (PIN) and the manufacturers name and part number (see 6.2).

3.6 Workmanship. Workmanship shall be such as to preclude evidence of careless handling, such as scratches and mars on exposed surfaces, blemishes, sharp edges, burrs, excessive brazing, and the like shall be cause for rejection (see 4.5.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspections (see 4.3).
c. Control tests (see 4.4).

4.2 First article inspection. First article inspection shall be performed on preproduction or initial production samples as specified when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.3.2 (see table I) and the tests of 4.5.4 through 4.5.4.6 (see table II) in the sequence shown in table III.

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TABLE I. Classification of defects.

Category	Defect	Method of examination
Major:		
101	Brazing, improper method (see 3.3).	Visual
102	Tubes not as specified (see 3.3).	Visual & SIE 1/
103	Corrosion and flux control - improperly removed (see 3.3.1).	Visual & Functional
104	Internal preservation treatment - insufficient protection (see 3.3.2).	Visual & Functional
105	Paint pretreatment - improperly performed (see 3.3.3).	Visual
106	Dimensions affecting interchangeability - out of tolerance (see 3.3.4).	SIE & Functional
107	Workmanship affecting performance - insufficient (see 3.6).	SIE & Functional
Minor:		
201	Painting - improper application or finish (see 3.3.3).	Visual
202	Dimensions not affecting interchangeability - out of tolerance (see 3.3.4).	SIE
203	Marking - illegible or improperly performed (see 3.5).	Visual
204	Workmanship - improperly performed (see 3.6).	Visual

1/ SIE = Standard Inspection Equipment.

TABLE II. Classification of inspections.

Title	Requirement	Inspection	First article	Conformance	Control test
Materials & construction	3.2 thru 3.3.2	4.5.1	X		
Defects (see table I)	3.3.3, 3.3.4, 3.5 & 3.6	4.5.2	X	X	X
Performance	3.4	4.5	X		
Leakage (noncyclic)	3.4.1	4.5.4.1	X	X	
Leakage (cyclic)	3.4.1	4.5.4.2	X	X	
Hydrostatic pressure	3.4.2	4.5.4.3	X		X
Heat rejection	3.4.3	4.5.4.4	X	X	X
Heat rejection, arrangement A	3.4.3.1	4.5.4.4.1	X		X
Heat rejection, arrangement B	3.4.3.2	4.5.4.4.2	X		X
Heat rejection, arrangement C	3.4.3.3	4.5.4.4.3	X		X
Vibration	3.4.4	4.5.4.5	X		X
Corrosion resistance	3.4.5	4.5.4.6	X		X

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TABLE III. First article test sequence.

Test sequence	Inspection	Core assembly, engine oil, configuration A	Core assembly, transmission oil, configuration B	Oil cooler assembly, configuration C
Leakage	4.5.4.1	X	X	
Cycling	4.5.4.2	X	X	
Vibration	4.5.4.5	X	X	
Corrosion	4.5.4.6	X	X	
Hydrostatic	4.5.4.3	X	X	X
Heat rejection, configuration A	4.5.4.4.1	X		
Heat rejection, configuration B	4.5.4.4.2		X	
Heat rejection, configuration C	4.5.4.4.3			X

4.3 Conformance inspections. Conformance inspections shall include the examinations of 4.3.2 (see table I) and the tests of 4.5.4 through 4.5.4.6 (see table II).

4.3.1 Sampling. Samples for conformance inspection shall be selected in accordance with ANSI/ASQC Z1.4.

4.3.2 Examination. Samples for examination inspection shall be in accordance with Table I.

4.3.3 Tests. Each sample shall be tested for compliance with the requirements specified in 3.4.1.

4.4 Control test. When specified (see 6.2), control test samples shall be selected at the rate of 2 assemblies from each 500 units of one configuration produced, except that not more than 4 or less than 2 samples shall be selected in any 30 day period. The control test samples shall be subjected to the examination and control tests specified in table II.

4.5 Methods of inspection.

4.5.1 Materials and construction. Conformance to 3.2 thru 3.3.2 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

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4.5.2 Defects. Conformance to 3.3 through 3.3.4, 3.5 and 3.6 shall be determined by examination for the defects listed in table I. Examination shall be visual, tactile, or by measurement with Standard Inspection Equipment (SIE).

4.5.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 73 ± 18 degrees Fahrenheit (°F).
b. Barometric pressure 28.5 (+2, - 3) inches of mercury (Hg).
c. Relative humidity 50 ± 30%.

4.5.4 Performance. Conditions for tests hereunder shall be as specified in 3.4.

4.5.4.1 Leakage tests. To determine conformance to 3.4.1, each sample item shall be submerged in water at temperature of 135 to 150°F. While submerged, an internal pressure, using air or nitrogen, shall be gradually applied until a pressure of 200 pounds per square inch (psi) has been reached. This pressure shall be maintained for 1 minute to determine conformance to 3.4.1.

4.5.4.2 Cycling. To determine conformance to 3.4.1, each sample item shall be submerged as specified in 4.5.4.1 and subjected to 10 cycles of air or nitrogen pressure with each cycle consisting of increasing the pressure from 0 to 200 psi and holding the maximum pressure for 15 seconds.

4.5.4.3 Hydrostatic test. To determine conformance to 3.4.2, each sample item shall be subjected to a hydrostatic pressure of 300 psi for 5 minutes. Cores subjected to this test shall not be acceptable as part of the quantity of the contract and are not to be shipped.

4.5.4.4 Heat rejection. To determine conformance to 3.4.3, each sample to be heat rejection tested shall be mounted in a test setup to produce the proper flow circuit for that core arrangement (see figure 1). Testing shall be conducted using SAE grade 10W oil.

4.5.4.4.1 Heat rejection test, core arrangement A. To determine conformance to 3.4.3.1, the oil entering the cores of engine core assemblies shall be at 215°F and at oil flow rates of 270, 204 and 135 pounds per minute (lb/min) with the air temperature to the core at 90 ± 10°F and the air flow rates of 80 to 180 lb/min per square feet (ft^2) in 20 lb/min per ft^2 increments (see figure 2). Testing shall be conducted using SAE grade 10W oil.

4.5.4.4.2 Heat rejection test, core arrangement B. To determine conformance to 3.4.3.2, the oil entering transmission oil core assembly shall be at 300°F and at oil flow rates of 375, 300, 200, and 100 lb/min with an air temperature of 90 ± 10°F to the core and air flow rates of 80 to 180 lb/min per ft^2 increments (see figure 3).

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4.5.4.4.3 Heat rejection test, engine and transmission oil core assemblies, core arrangement C. To determine conformance to 3.4.3.3, the oil entering the engine and transmission oil core assemblies shall be at 300°F and at flow rates of 375, 300, 200, and 100 lb/min with an air temperature of 90 ± 10°F to the core and at air flow rates of 80 to 180 lb/min per ft^2 in 20 lb/min per ft^2 increments (see figure 4).

4.5.4.5 Vibration test. To determine conformance to 3.4.4, the sample core assembly shall be tested as specified below.

4.5.4.5.1 Test set-up. The sample core assembly shall be suspended vertically, bottom end up, by means of two flexible cords (suggested aircraft shock cord) attached by brackets A and B to the cooler mounting angles as shown in figure 9. With the core assembly suspended in this manner, vibration shall be applied in a horizontal direction to one of the lower corners with the applied vibration perpendicular to the face of the core. The vibration shall be applied to the corner of the core assembly by means of block C as shown in figure 10. Bracket B, which shall be constructed for mounting a vibration pick-up, and block C shall be so arranged as to be on diagonally opposite corners of the core. Vibration pickups shall then be attached to bracket B and block C, as shown in figures 9 and 10, for measuring vibration in a horizontal direction perpendicular to the face of the core assembly. The core assembly shall be filled with SAE grade 10W oil and subjected to a static pressure of 100 psi during the test by use of air pressure or other means.

4.5.4.5.2 Resonant frequency survey. With the core set-up as specified in 4.5.4.5.1, vibration shall be applied and the frequency varied until a resonant torsional mode of vibration has been found of 65 to 70 cycles per second (cps). Mode of vibration shall produce nodal patterns as shown in figure 11. The peak to peak amplitude at the point of vibration shall not exceed 0.010 inch (in.) during the survey.

4.5.4.5.3 Test amplitudes. With the core assembly vibrating at the resonant frequency, the double amplitude at the point of vibration application shall be increased to 0.0245 in. at which time the amplitude at bracket B should be 0.0340 in. Specified amplitudes shall be maintained within 3%. If the amplitude at the latter point is slightly different than the specified value, adjustment shall be made as follows:

a. An increase in the amplitude at bracket B may be obtained by a slight decrease in the frequency while still maintaining the same amplitude at the point of vibration application at block C. A decrease in the amplitude at bracket B may be obtained by a slight increase in the frequency, while still maintaining the same amplitude at the point of vibration application, block C.
b. When originally setting up the specified amplitudes, the amplitudes should be increased slowly so as to not exceed the specified amplitudes. Setting the

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amplitude at block C about 20% low will allow the operator to judge whether the amplitude at bracket B is going to be correct or not, at which time slight frequency adjustments may be made.

4.5.4.5.4 Test procedure. To determine conformance to 3.4.1, the core assembly shall be vibrated for 2 000 000 vibration cycles. After completion of vibration, the core assembly shall be subjected to the test specified in 4.5.4.2. Evidence of leakage during any stage of the test shall be considered as failure to pass the test. Cores subjected to this test shall not be acceptable as part of the quantity of the contract and are not to be shipped.

4.5.4.6 Corrosion resistance test. To determine conformance to 3.4.5, oil cooler assemblies or core assemblies, with end plugged, shall be subjected to a 336-hour salt spray test in accordance with ASTM B117.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The oil cooler assemblies covered by this specification are intended for use in military tactical and transport vehicles to cool the vehicle engine and transmission oil.

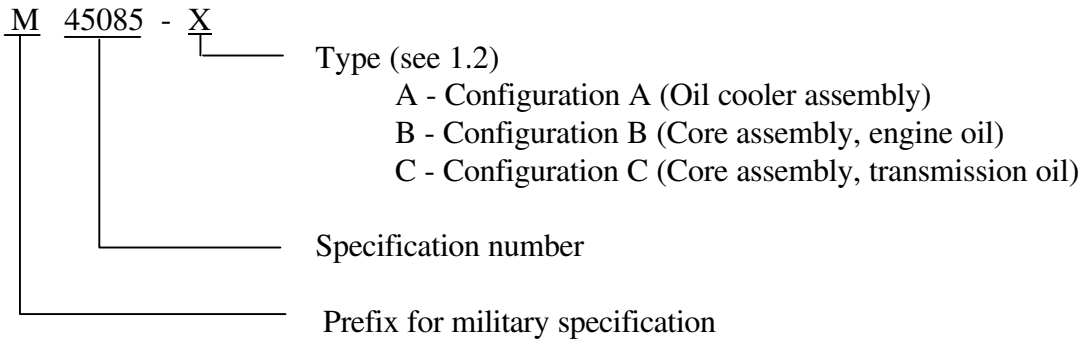
6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Classification of Assembly/ PIN (see 1.2, 3.5 and 6.3).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. When first article, is not required (see 3.1).

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e. Control test, if required (see 4.4).
f. Inspection conditions, if other than specified (see 4.5.3).
g. Packaging requirements (see 5.1).

6.3 Part or Identification Number (PIN). The PIN to be used for oil cooler assemblies acquired to this specification are created as follows:

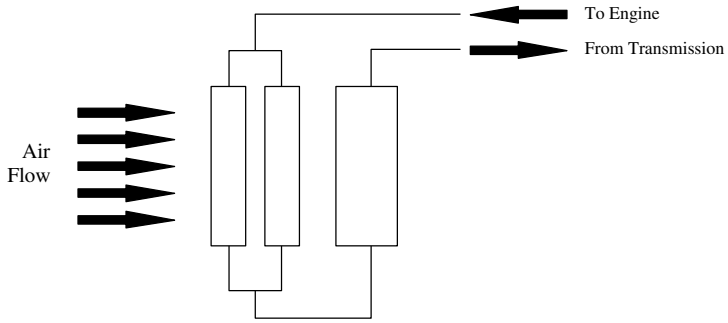


6.4 Subject term (key word) listing.

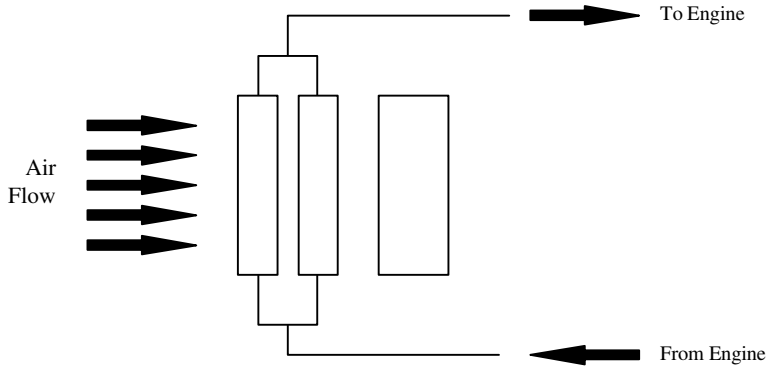
Core
Fluid
Temperature

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

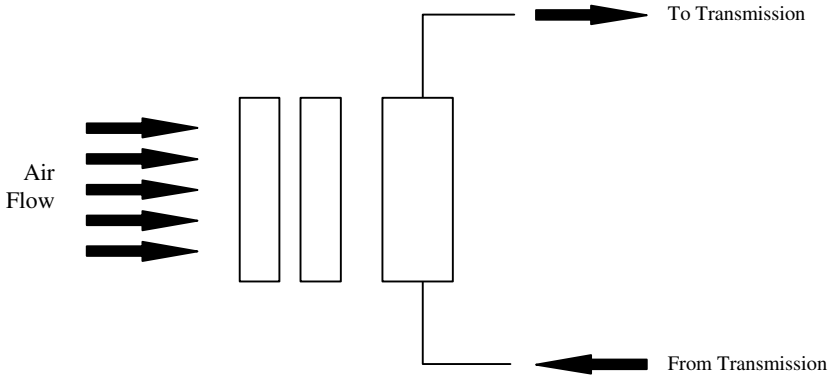
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CORE ARRANGEMENT A. ENGINE & TRANSMISSION (DRW.8690239)



CORE ARRANGEMENT B. ENGINE CORE (DRW.7356770)



CORE ARRANGEMENT C. TRANSMISSION CORE (DRW.7356769)

NOTE: All tests made with air flowing through three cores (two engine cores and one transmission core).

FIGURE 1. Flow circuit of core arrangements.

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CORE ARRANGEMENT A. ENGINE OIL IN-TEMPERATURE 215°F.

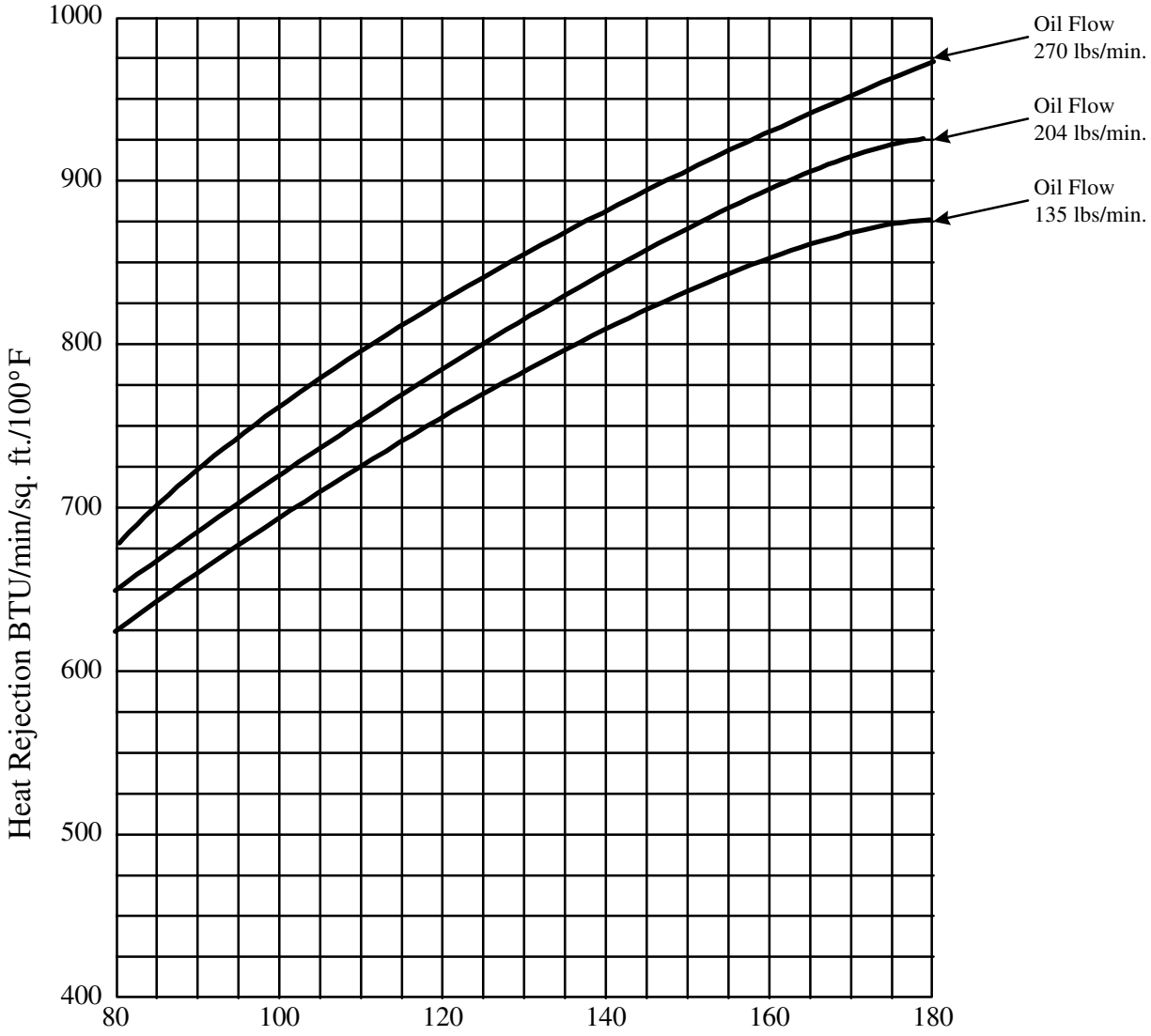


FIGURE 2. Cooling performance.

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CORE ARRANGEMENT B. TRANSMISSION OIL IN-TEMPERATURE 300°F.

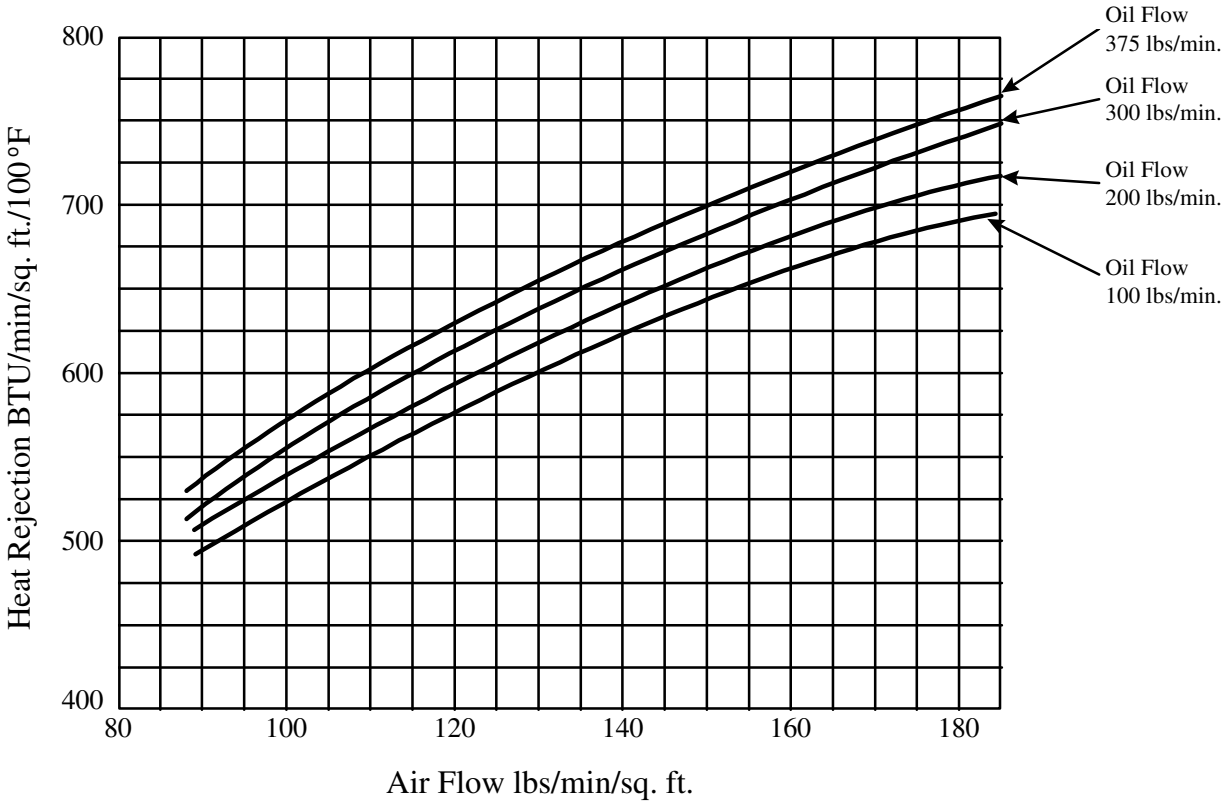


FIGURE 3. Cooling performance.

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CORE ARRANGEMENT C. COMBINED OIL IN-TEMPERATURE 300°F.

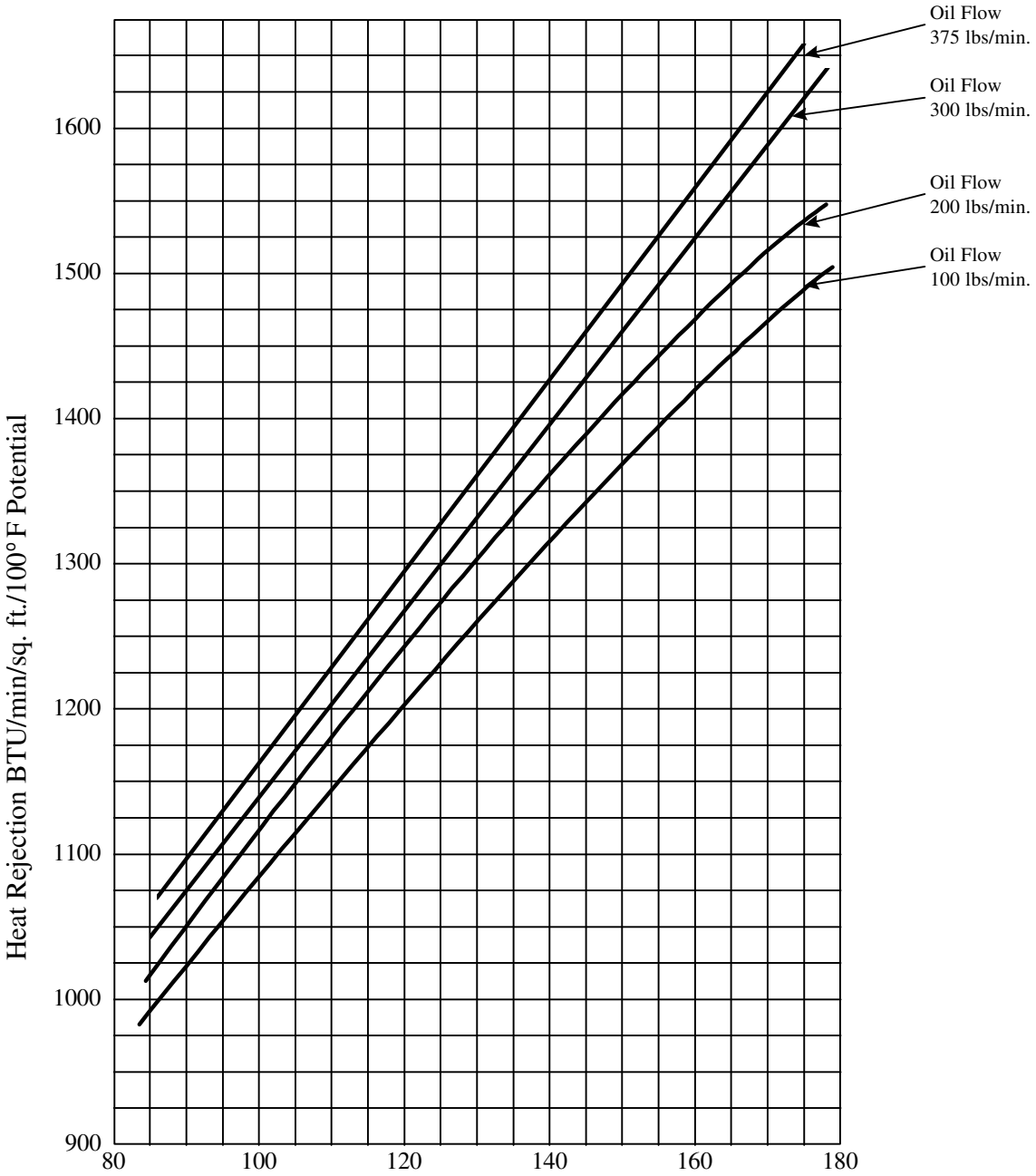


FIGURE 4. Cooling performance.

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CORE ARRANGEMENT A. ENGINE OIL IN-TEMPERATURE 215°F.

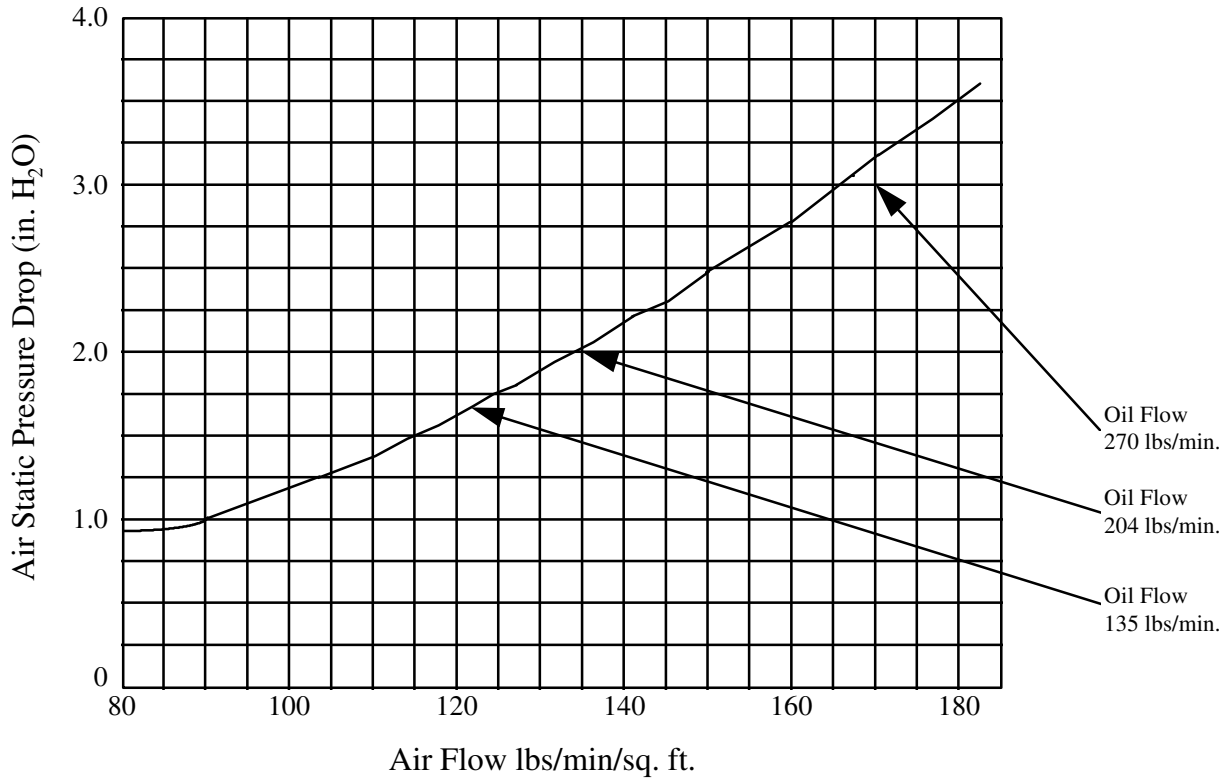


FIGURE 5. Air flow characteristics.

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CORE ARRANGEMENT B. TRANSMISSION OIL IN-TEMPERATURE 300°F.

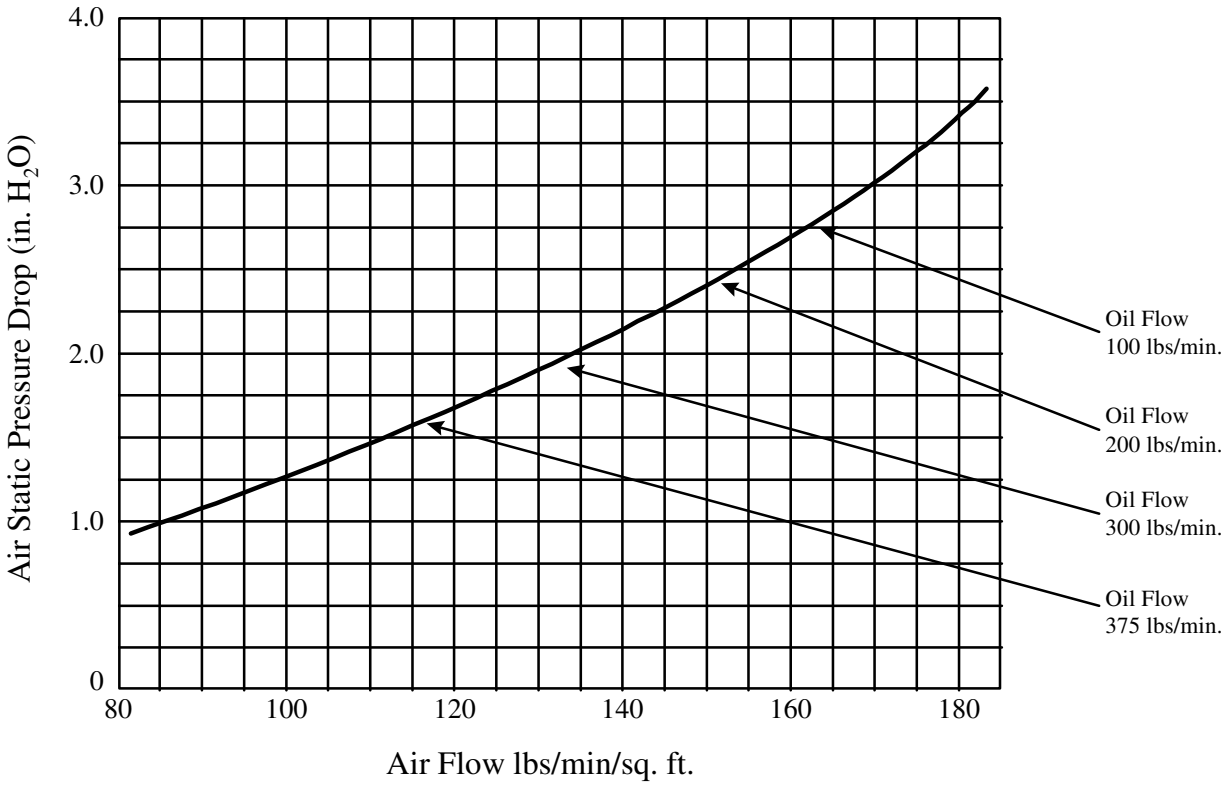


FIGURE 6. Air flow characteristics.

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CORE ARRANGEMENT C. TRANSMISSION OIL IN-TEMPERATURE 300°F.

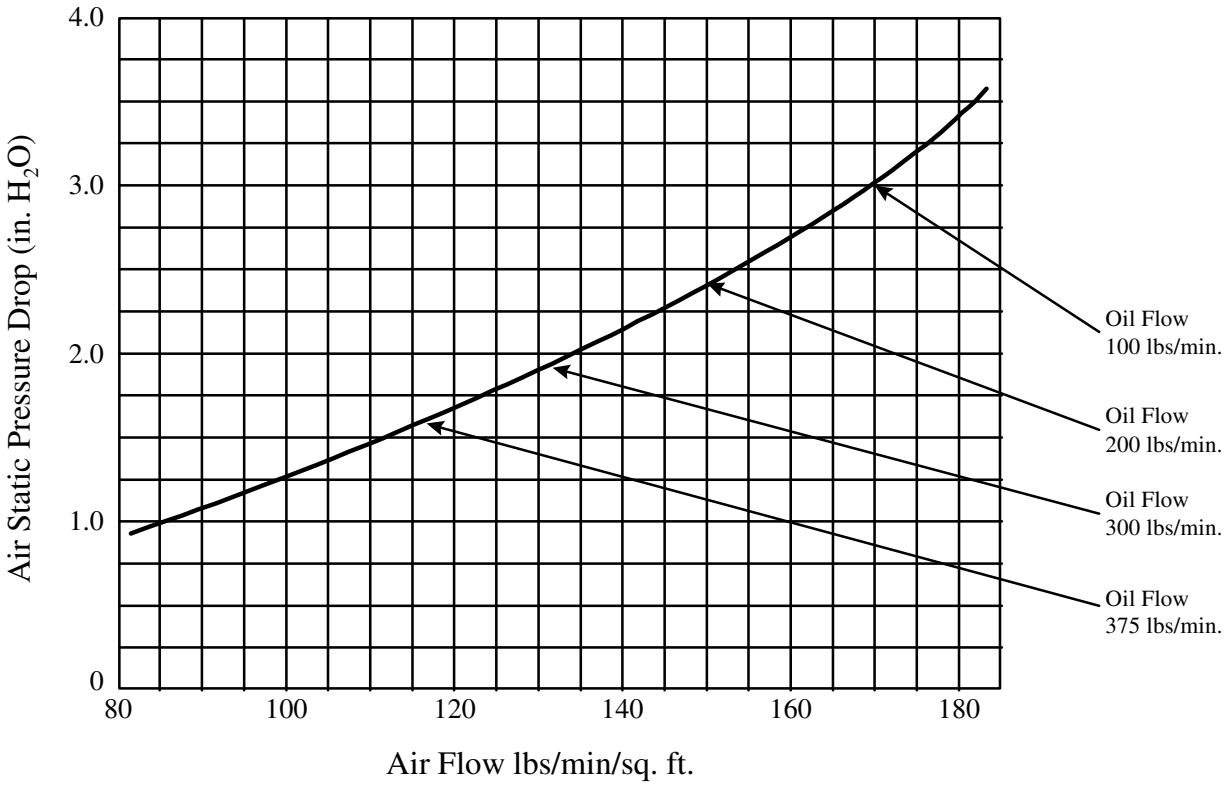


FIGURE 7. Air flow characteristics.

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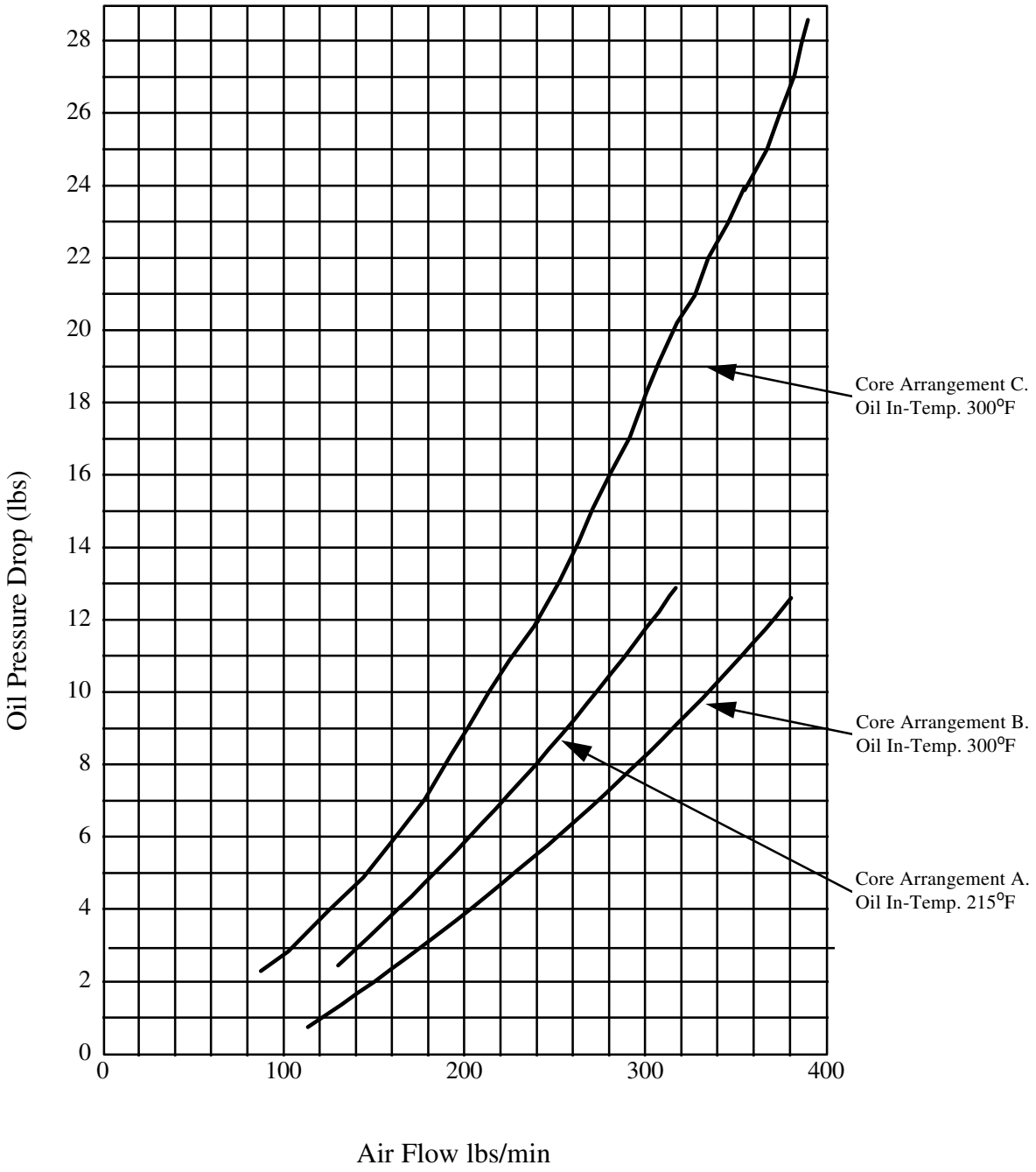


FIGURE 8. Oil pressure drop.

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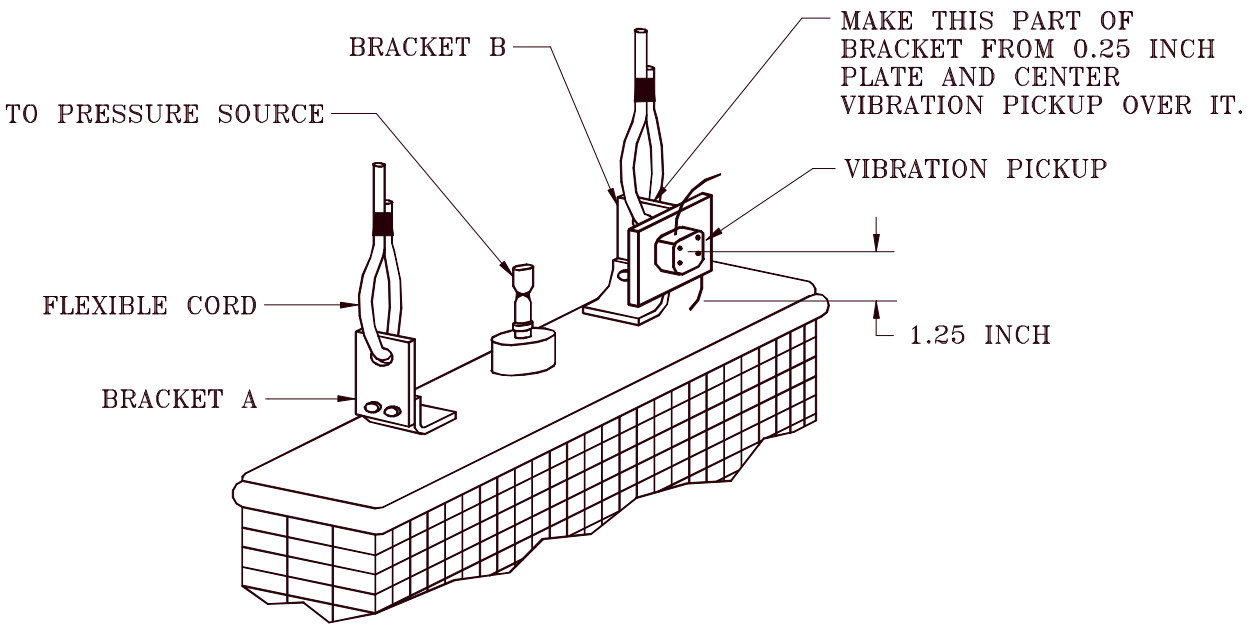


FIGURE 9. Cooler test setup, bracket assembly.

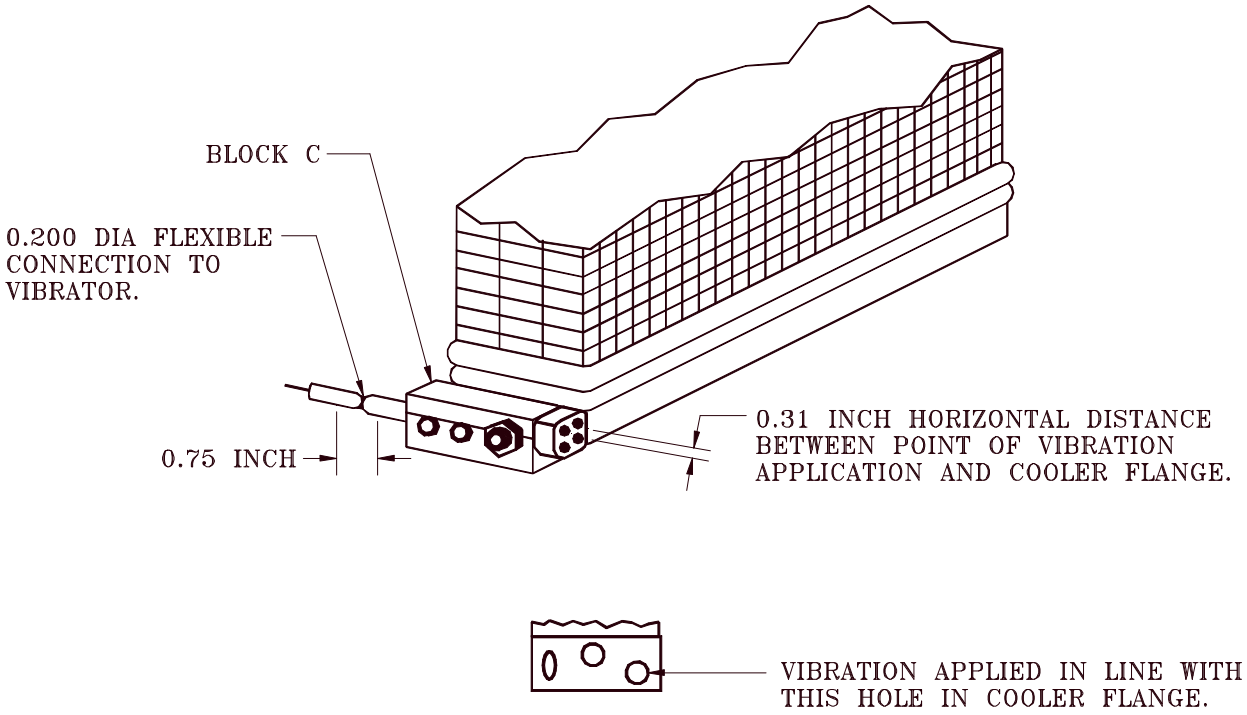
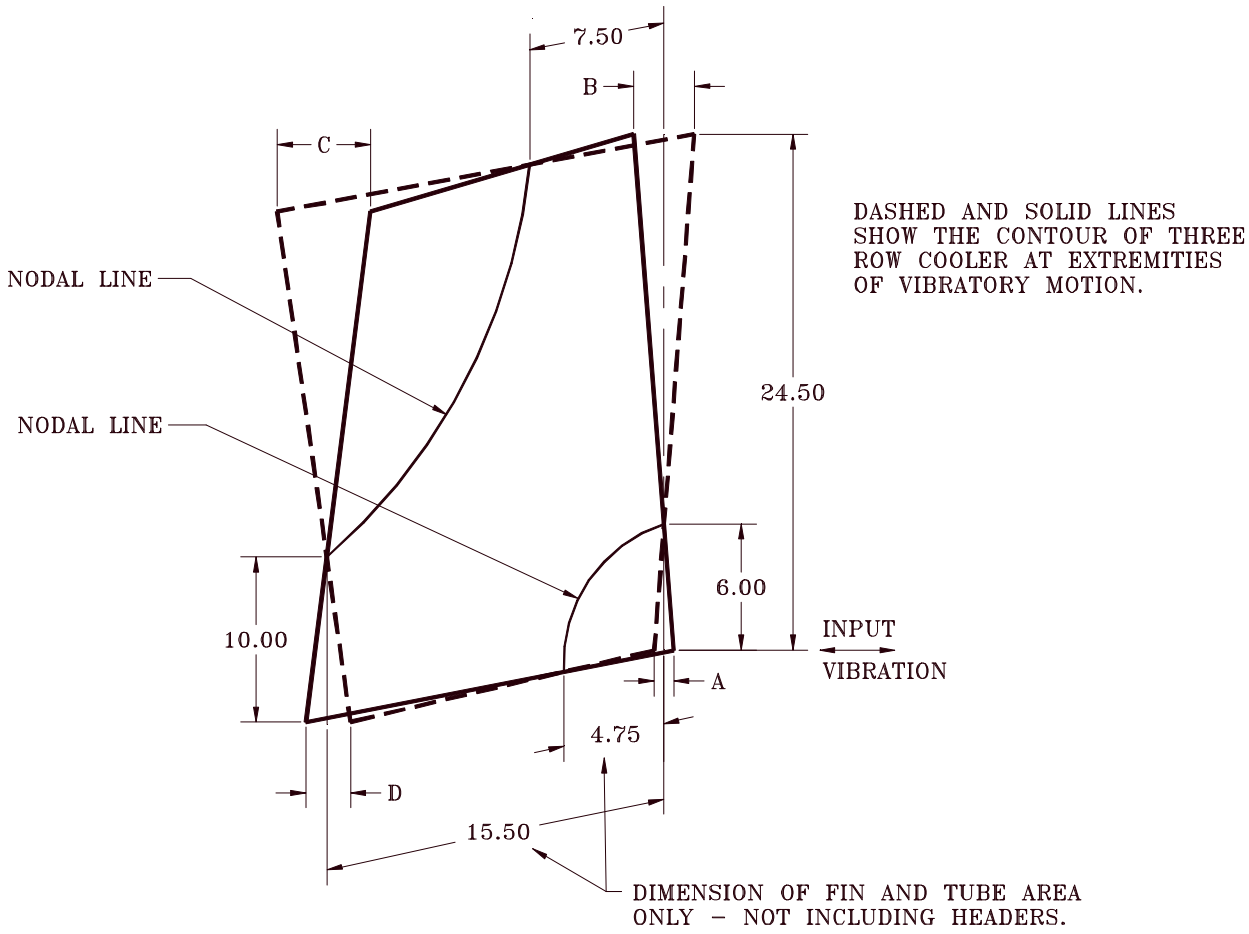


FIGURE 10. Cooler test setup, corner assembly.

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NOTE: DIMENSIONS IN INCHES.

FIGURE 11. Vibration mode used in endurance tests of oil coolers.

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3. DOCUMENT TITLE
Oil Cooler Assembly: Engine and Transmission

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)***

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